

- 3 (a) As the body moves around the circle with constant speed, there is a (continuous) change in velocity due to a (continuous) change in direction and hence the body has an acceleration. **B1**

Since the rate of change of velocity must be towards the centre, the acceleration must be directed towards the centre of the circle. **B1**

By Newton's second law of motion (resultant force), there must be a force acting on the object in the direction of the acceleration. Hence a body moving with uniform speed in a circle must experience a force towards the centre of the circle and perpendicular to the motion. **B1**

- (b) (i) Normal contact force acting on the man by the spacecraft. **B1**
*do not allow contact force

(ii) Since the person is in uniform circular motion,
 $a = r\omega^2$, where ω is the angular velocity in rad s^{-1} . **M1**

Therefore, $k = r\left(\frac{2\pi}{60}f\right)^2$ (shown) **M1**

$= 0.0110rf^2$ **A1**

*do not accept answer in π

(iii) $a = 9.81 \text{ m s}^{-2}$ **M1**

$9.81 = 0.0110r(2)^2$ **M1**

$r = 223 \text{ m}$ (to 3 sig. fig) **A1**